

What it does

A phone records a cricket delivery; user taps pitch corners and stump rectangles; system returns **OUT** / **NOT OUT** / **UMPIRES CALL** with a 3D Hawk-Eye trajectory.

Pipeline (5 stages)

1. **Calibrate** — solvePnP on 4 pitch corners + 8 stump-quad points; jointly auto-fits FOV (28–86°) and pitch length.
2. **Detect** — MOG2 motion + HSV colour + ROI mask, with Ultralytics YOLO cricket-ball fallback.
3. **Link** — RANSAC over constant-acceleration motion model; outputs (u, v, t) image-space track.
4. **Reconstruct 3D** — depth from ball size $d = f_x \cdot R_{\text{real}} / r_{\text{px}}$ ($R_{\text{real}} = 0.036$ m); projectile-fit with $g = 9.81$ m/s² + bounce reflection (Ribnick 2009 linear solve).
5. **Decide** — ICC Rule 36: pitched in line + impact in line + hitting stumps; 25 mm umpire's-call margin.

Key formulas

- Pinhole: $u = f_x X / Z + c_x$, $v = f_y Y / Z + c_y$
- f_x from FOV: $f_x = (\max(W, H) / 2) / \tan(\text{FOV} / 2)$
- Depth from ball: $d_m = f_x \cdot 0.036 / r_{\text{px}}$
- Projectile: $z(t) = z_0 + v_z t - \frac{1}{2} g t^2$ ($g = 9.81$)
- Bounce reflection: $v_{z, \text{post}} = -e v_{z, \text{pre}}$ ($e \approx 0.55$)

ICC pitch geometry

- Pitch: 20.12 m × 3.05 m
- Stumps: 3 stumps, 0.711 m tall, 0.114 m apart (centre-to-centre); 0.018 m thick $\Rightarrow$ outermost edge ± 0.132 m
- Bails: 0.012 m above stumps
- LBW in-line corridor: outer edge ± 0.036 m (ball radius)

Decision rules (Rule 36)

- Pitched in line of stumps? (inside corridor)
- Impact in line of stumps? (offside-of-bat exception *not* implemented)
- Hitting stumps? (path intersects stump volume + 25 mm band)
- **OUT** only if all three pass with margin > 25 mm
- Margin within 25 mm $\Rightarrow$ **UMPIRES CALL**
- Any check fails $\Rightarrow$ **NOT OUT**

Real-video results (2026-05-24)

- **test3.mp4** (zoomed phone, FOV 37°): 64.1 km/h, decision **NOT OUT** (predicted ball misses stumps); reproj 16.6 px, 28 inliers.

- Real-phone HEVC truncation handled by pipeline truncation guard.

Synthetic validation (8 scenarios)

- **8/8 PASS** at honest monocular bounds: ± 55 km/h speed, ± 175 cm bounce, ± 110 cm impact.
- Mean errors: speed 26.7 km/h, bounce 60 cm, impact 63 cm.
- Sweet-spot setup: 115 km/h, middle corridor, 5.5–6 m length, full ICC pitch.
- Pipeline rejects fits beyond bounds — never emits confident-wrong decisions.

Stack

- Flutter 3.8 (Dart) mobile, FastAPI 0.104 + Python 3.11 backend.
- OpenCV 4.8, NumPy 1.24, SciPy 1.11, Ultralytics YOLO.
- Three.js for 3D viewer.
- Firebase Auth + Firestore + Storage + Cloud Run.
- Free-tier hostable; one phone is the only hardware.

Hardening

- Calibration rejected if reproj > 8 px (large-error pose = meaningless physics).
- 3D fit rejected if RMS > 1.0 m (untrustworthy).
- VideoReader uses frame-index seeking (phone HEVC sparse keyframes).
- Truncation guard stops at last good frame (no stale-frame padding).
- Camera height > 0 invariant prevents physically impossible solves.

Key references

- Owens et al. 2003 — original Hawk-Eye (VIE).
- Zhang 2000 — checkerboard calibration (IEEE TPAMI).
- Fischler & Bolles 1981 — RANSAC (CACM).
- Hartley & Zisserman 2004 — Multi-View Geometry (book).
- Ponglertnapakorn & Suwajanakorn 2025 — Monocular 3D ball tracking (CVPRW).
- ICC Rule 36 — LBW.

Defense Q&A

- **How accurate?** ~ 6 cm bounce, ~ 80 cm impact, ± 25 km/h speed monocular. Real Hawk-Eye = sub-cm with 6 cameras.
- **Why not multi-camera?** Phone-only = affordable for amateur cricket; trade accuracy for accessibility.
- **What if calibration is wrong?** Pipeline rejects (reproj > 8 px $\Rightarrow$ error); no confident-wrong outputs.